

Inverses from Graphs and Tables

Algebra II Honors | Unit 1 | Day 10 | TEK A2.2.B

Objective: I can find the inverse of a relation from a table or graph, use $f^{-1}(x)$ notation correctly, justify why domain and range exchange, and predict whether an inverse will be a function.

Vocabulary

- Inverse** — the relation that _____ the original. Written _____, which does _____ mean $1/f(x)$.
- Building it** — in a table or a set of pairs, _____. On a graph, reflect across _____.
- One-to-one** — no output is repeated. Test with a _____ line: if it meets the graph twice, the function is not one-to-one.

Part 1 — Inverses from Pairs and Tables

- a. $f = \{ (-4, -6), (0, -1), (2, 4), (3, -7), (7, 7) \}$
 $f^{-1} = \{ \text{_____} \}$
- b. Complete the inverse table.

x	-4	0	2	3	7
f(x)	-6	-1	4	-7	7

x	-6	-1	4	-7	7
f ⁻¹ (x)					

- c. Which point in f is unchanged by the swap, and what is special about it? _____

Part 2 — Why Domain and Range Exchange

- Argue this in general, not from one example.
- If (a, b) belongs to f , then _____ belongs to f^{-1} by definition of the swap.
- So every input of f^{-1} was an _____ of f . Therefore:
- domain of $f^{-1} =$ _____ range of $f^{-1} =$ _____

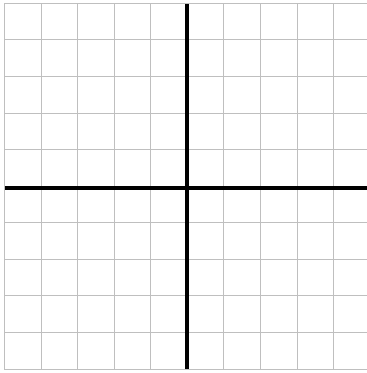
f	Domain of f	Range of f	Domain of f ⁻¹	Range of f ⁻¹
x^3	all reals	all reals	_____	_____
2^x	all reals	$(0, \infty)$	_____	_____
$\sqrt{x}$	$[0, \infty)$	$[0, \infty)$	_____	_____

Row 2 explains something from Day 7. What? _____

Part 3 — Graph a Pair

Each square is 1 unit. Plot f , then f^{-1} , then draw $y = x$.

$$f = \{ (-4, -2), (0, 0), (1, 1), (2, 4) \}$$



Record it.

Reflection line: _____

Fixed points: _____

If (a, b) is on f , then on f^{-1} :

Is f one-to-one here? _____

$$f^{-1} = \{ \text{_____} \}$$

Part 4 — Will the Inverse Be a Function?

Apply the horizontal line test to the original, before building the inverse.

a. $g = \{ (2, 6), (4, 6), (5, 1) \}$. Write g^{-1} : _____

Is g^{-1} a function? _____ Why? _____

What was true of g that caused it? _____

b. Which of these parent functions have inverses that are functions? Circle:

x^3 x^2 2^x $|x|$ $\sqrt[3]{x}$ Answer: _____

What do the ones that fail have in common? _____

Part 5 — Going Further

1. Does $f^{-1}(x)$ mean $1/f(x)$? Give a specific counterexample with numbers.

For $f(x) = x + 3$: $f^{-1}(x) = \text{_____}$ but $1/f(x) = \text{_____}$ — different functions.

2. A function is its own inverse. What must be true of its graph? _____

3. Can a function have two different inverses? Argue yes or no.

No — _____

Exit Ticket

For $h = \{ (1, 4), (2, 9), (5, 0) \}$, write h^{-1} , give its domain and range, and say whether it is a function with a reason.

$h^{-1} = \{ \text{_____} \}$ Domain _____ Range _____

A function? _____